

CD25 Deficiency (IL2RA Deficiency): A Comprehensive Disease Characteristics Report

Disease Name: CD25 Deficiency (Immunodeficiency 41 with lymphoproliferation and autoimmunity, IMD41) **MONDO ID:** MONDO:0011664 **Category:** Mendelian (monogenic inborn error of immunity)

Summary

CD25 deficiency is an **ultra-rare autosomal recessive inborn error of immunity** caused by biallelic loss-of-function variants in **IL2RA** (chromosome 10p15.1), the gene encoding CD25, the alpha chain of the high-affinity interleukin-2 (IL-2) receptor. CD25, in complex with CD122 (IL-2R β) and the common gamma chain (γ c), confers high-affinity IL-2 binding. Loss of this chain abolishes high-affinity IL-2/JAK1-JAK3/STAT5 signaling, which is indispensable for the development, survival, stability, and suppressive function of FOXP3⁺ regulatory T cells (Tregs) and for normal IL-10 production. The resulting failure of peripheral immune tolerance produces an **IPEX-like syndrome** that paradoxically combines features of immunodeficiency (recurrent viral, bacterial, and fungal infections) with early-childhood **autoimmunity and lymphoproliferation** (autoimmune enteropathy, hepatitis, thyroiditis, cytopenias, eczema, multi-organ lymphocytic infiltration, and severe failure to thrive).

The disorder was first described by Sharfe et al. in 1997 as a truncation mutation of the IL-2 receptor alpha chain producing profound cellular immunodeficiency with extensive lymphocytic tissue infiltration. Fewer than ~15 molecularly confirmed cases have been reported worldwide as of 2026, so essentially all knowledge derives from **individual patient case reports** rather than aggregated registries; no population prevalence or incidence estimate is established. Diagnosis rests on demonstrating **absent surface CD25** on T cells by flow cytometry, followed by **IL2RA sequencing** for molecular confirmation. The differential diagnosis centers on IPEX (FOXP3) and other IPEX-like Tregopathies.

Without definitive treatment the disease is severe and often fatal in early childhood from overwhelming infection or autoimmune organ damage. **Allogeneic hematopoietic stem cell transplantation (HSCT)** is the only curative therapy, restoring functional Tregs, and can produce complete resolution of symptoms. **Rapamycin (sirolimus)**, an mTOR inhibitor that favors Tregs, has been used as effective and safe bridging immunomodulation. Because the disorder is Mendelian, prevention is limited to genetic counseling, cascade carrier testing (especially in consanguineous families), and prenatal/preimplantation genetic testing once the familial variant is known. Mouse models (Il2^{-/-} and Il2ra/CD25 knockout) recapitulate the human phenotype and demonstrate genetic-background-dependent modifier effects.

1. Disease Information

Overview. CD25 deficiency is a monogenic, autosomal recessive inborn error of immunity in which loss of the IL-2 receptor alpha chain (CD25) disables high-affinity IL-2 signaling and thereby cripples regulatory T-cell function. The clinical picture is described as "IPEX-like" because it phenocopies IPEX (Immune dysregulation, Polyendocrinopathy, Enteropathy, X-linked syndrome) caused by FOXP3 mutations, but with a normal FOXP3 gene. Sharfe et al. (1997) described "a novel human immune aberration arising from a truncation mutation of the interleukin-2 receptor alpha chain (CD25)... characterized by decreased numbers of peripheral T cells displaying abnormal proliferation but normal B cell development. Extensive lymphocytic infiltration of tissues, including lung, liver, gut, and bone, is observed, accompanied by tissue atrophy and inflammation" ([PMID: 9096364](#)).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0011664
OMIM (phenotype)	#606367 — IMMUNODEFICIENCY 41 WITH LYMPHOPROLIFERATION AND AUTOIMMUNITY (IMD41)
OMIM (gene)	*147730 (IL2RA)
Orphanet	ORPHA:169153 — Immunodeficiency due to a defect in CD25
HGNC (gene)	HGNC:6008 (IL2RA)
NCBI Gene	3559
UniProt	P01589
Chromosome	10p15.1
ICD-10	D81.8 / D84.9 (immunodeficiency range)
MeSH	via "Interleukin-2 Receptor alpha Subunit"

Synonyms / alternative names. IL2RA deficiency; interleukin-2 receptor alpha chain deficiency; CD25 deficiency syndrome; Immunodeficiency 41 with lymphoproliferation and autoimmunity (IMD41); IPEX-like syndrome due to CD25 deficiency.

Data provenance. Because only ~13 molecularly confirmed cases had been reported worldwide as of 2026 ([PMID: 41659858](#)), disease-level knowledge derives almost entirely from **individual patient case reports and small case series**, not from aggregated EHR/registry resources.

2. Etiology

Causal factors. CD25 deficiency is **purely genetic** — caused by biallelic (homozygous or compound heterozygous) germline loss-of-function variants in *IL2RA*. There is no environmental, toxic, occupational, or lifestyle cause. Loss-of-function IL2RA variants "cause a very rare autosomal recessive disorder marked by early-onset autoimmunity and recurrent infections with an IPEX-like presentation" ([PMID: 41694357](#)).

Genetic risk factors. The only causal risk factor is inheriting two defective *IL2RA* alleles. The principal **population risk factor is parental consanguinity** — reported patients are frequently born to first-cousin parents ("Both patients were born to first-cousin parents," [PMID: 41694357](#)), consistent with a recessive disorder concentrated in inbred pedigrees.

Environmental / lifestyle risk factors. None are causal. Infections act as **triggers and complications** of the underlying immunodeficiency rather than as causes.

Protective factors. No genetic or environmental protective factors are established. Because the disease is Mendelian and fully penetrant when biallelic LOF variants are present, "protection" is effectively the absence of a second pathogenic allele (carriers are healthy).

Gene–environment interactions. In humans these are not well characterized owing to case rarity. However, mouse models demonstrate a clear **modifier / gene–environment interaction**: *Il2*^{-/-} mice develop ulcerative-colitis-like disease on a mixed 129/Ola × C57BL/6 background but generalized systemic autoimmunity when backcrossed to BALB/c, showing that genetic background dictates the phenotype ([PMID: 9065030](#)). Gut microbial antigens are inferred to drive the autoimmune enteropathy.

Important distinction — Mendelian LOF disease vs. common autoimmune-susceptibility polymorphisms. The rare Mendelian CD25 deficiency is mechanistically distinct from the well-established role of **common non-coding IL2RA regulatory polymorphisms** (e.g., rs2104286, rs12722495/rs12722496, rs41295061, rs7093069) as **polygenic susceptibility loci** for type 1 diabetes, multiple sclerosis, and autoimmune thyroid disease. Fine-mapping identified multiple independent T1D and MS association signals in the *IL2RA*/CD25 region ([PMID: 26106896](#)); *IL2RA*-rs41295061 (10p15) association was replicated in T1D ([PMID: 21875375](#)); and *IL-2RA* rs7093069 (TT genotype) was associated with pediatric autoimmune thyroid disease ([PMID: 33193078](#)). These common variants subtly tune CD25 expression and confer complex-trait autoimmune risk, whereas the rare biallelic LOF variants abolish CD25 entirely and cause the monogenic syndrome.

3. Phenotypes

Phenotypes are curated from reported cases (PMID: [9096364](#), [17196245](#), [23416241](#), [24116927](#), [29252577](#), [30742970](#), [41694357](#), [41659858](#)). Because fewer than ~15 cases exist, frequencies are **qualitative** (most/common) rather than precise percentages. Onset is typically **neonatal to early childhood** (congenital/pediatric); severity is **severe**; the course is **chronic and progressive** without treatment.

Phenotype	Type	Suggested HPO term	Frequency (qualitative)
Chronic diarrhea / autoimmune (celiac-like) enteropathy	Sign / GI	HP:0002028; HP:0002590	Majority
Failure to thrive / growth failure	Physical	HP:0001508	Majority
Recurrent respiratory infections	Sign	HP:0002205	Majority
Recurrent bacterial/viral/fungal infections	Sign	HP:0002719; HP:0002841	Majority
Eczema / dermatitis	Physical	HP:0000964	Common
Autoimmune hepatitis	Lab/ clinical	HP:0002608	Common
Autoimmune thyroiditis	Lab/ clinical	HP:0100646	Reported (≥ 1 case)
Autoimmune hemolytic anemia / cytopenias	Lab	HP:0001890	Reported
Lymphadenopathy / lymphoproliferation	Sign	HP:0002716	Common
Hepatosplenomegaly	Sign	HP:0001433	Common
Chronic inflammatory lung disease / follicular bronchiolitis	Clinical	HP:0006538	Reported
Keratitis / severe dry eye (ocular surface disease)	Clinical	HP:0000491; HP:0000492	Reported
Impaired T-cell proliferation	Lab	HP:0002850	Majority
Hypergammaglobulinemia	Lab	HP:0010702	Common
Absent CD25 surface expression	Lab	(diagnostic)	Universal

Key supporting quotes: patients "presented in early childhood with recurrent respiratory and gastrointestinal infections, severe failure to thrive, chronic diarrhea with celiac-like enteropathy, and autoimmune manifestations including autoimmune hepatitis, dermatitis, and, in one case, autoimmune thyroiditis" ([PMID: 41694357](#)); "Recurrent infections and

lymphocyte infiltration of multiple tissues are the main clinical presentations" (PMID: 29252577). Primary biliary cirrhosis has also been reported, notably as a rare pediatric occurrence (PMID: 24116927, PMID: 20650610).

Quality of life. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disease. Qualitatively, untreated disease imposes profound impairment: chronic diarrhea and malabsorption, failure to thrive, repeated hospitalizations for infection, autoimmune organ damage, and severe dry-eye discomfort all severely reduce daily functioning. Successful HSCT can restore near-normal functioning.

4. Genetic / Molecular Information

Causal gene. *IL2RA* (HGNC:6008; OMIM 147730; NCBI Gene 3559; UniProt P01589), located on chromosome 10p15.1*, encoding CD25, the 55-kDa alpha chain of the IL-2 receptor.

Pathogenic variants. Reported disease-causing variants are biallelic and loss-of-function, spanning: - **Nonsense/truncation** variants (the original Sharfe et al. 1997 case, PMID: 9096364). - **Missense** variants causing conformational loss of surface expression — e.g., c.122A>C, p.Tyr41Ser (Y41S), a homozygous missense mutation with no CD25 on CD4⁺ T cells and extremely low Tregs (PMID: 24116927); and a conformational mutation described as "a severe protein conformational alteration that abrogates its cell surface expression" (PMID: 30742970).

Variant classification. Reported variants are classified **pathogenic/likely pathogenic** per ACMG/AMP criteria (functional evidence of absent surface expression, segregation in consanguineous families, absence/rarity in population databases).

Allele frequency. Causal LOF variants are private/ultra-rare and essentially absent from gnomAD at appreciable frequency. By contrast, common regulatory SNPs in the locus are polymorphic (see Section 2).

Origin. **Germline**, biallelic. No somatic mechanism.

Functional consequence. **Loss of function** — loss of high-affinity IL-2 binding. CD25 "contributes only to IL-2 binding affinity but not to the recruitment of signalling molecules" (PMID: 24116927), so its loss reduces IL-2 receptor affinity rather than eliminating all IL-2 signaling capacity, but this is sufficient to cripple Treg biology.

Modifier genes. Not defined in humans. Mouse background effects (129 vs BALB/c) demonstrate strong modifier influence ([PMID: 9065030](#)). Related Treg biology implicates FOXP1/FOXP4, which bind Il2ra promoter regions to regulate CD25 expression ([PMID: 40794436](#)).

Epigenetic information. No disease-specific methylation/histone data. FOXP1/FOXP4 transcriptional control of the Il2ra promoter is the most relevant regulatory layer ([PMID: 40794436](#)).

Chromosomal abnormalities. None — this is a single-gene disorder, not a structural/aneuploidy syndrome.

5. Environmental Information

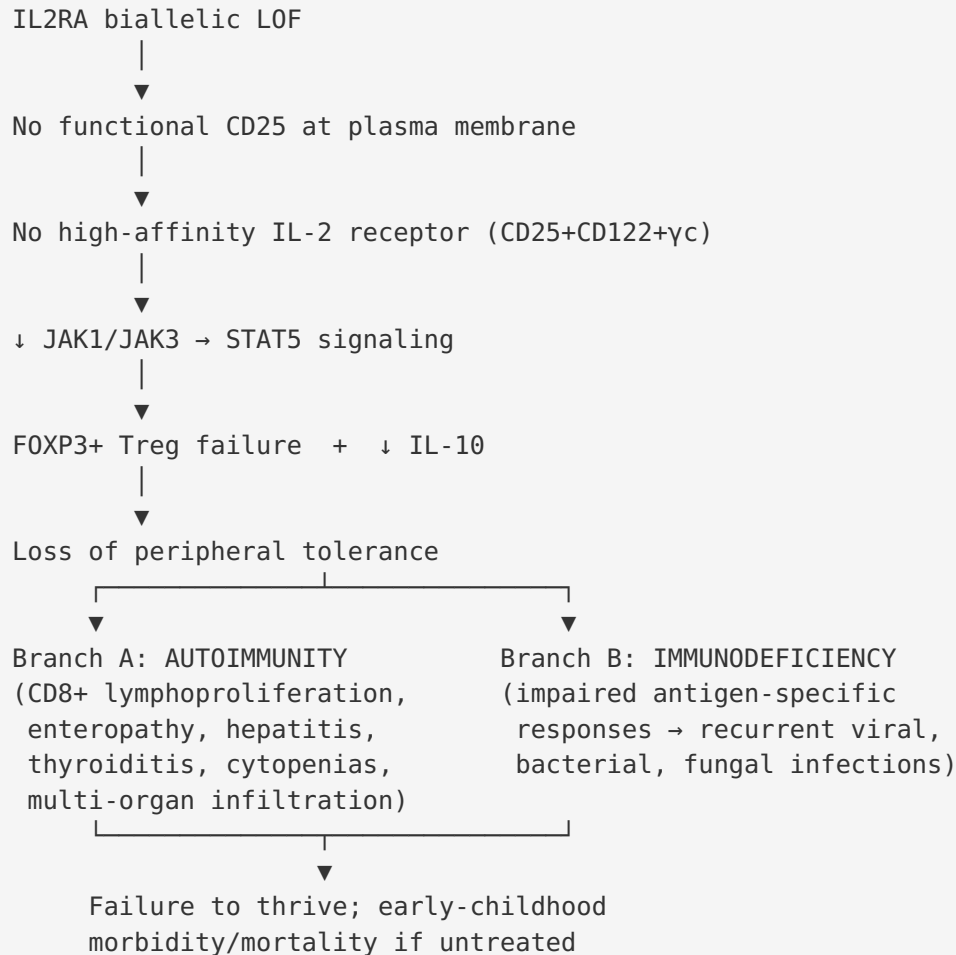
- **Environmental factors:** None causal. No toxin, radiation, pollution, or occupational exposure contributes to disease onset.
- **Lifestyle factors:** Not applicable (disease presents in infancy).
- **Infectious agents:** Pathogens act as **triggers/complications**, not causes. Patients suffer recurrent **viral (including CMV/herpesviruses), bacterial, and fungal (Candida)** infections due to the immunodeficiency. Gut microbial antigens are inferred to drive autoimmune enteropathy (by analogy to IL-2-pathway mouse colitis, [PMID: 9065030](#)).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in IL2RA** (nonsense/truncation or conformational missense) **lead to** absent or non-functional CD25 protein at the T-cell plasma membrane ([PMID: 9096364, 30742970](#)).
2. Loss of CD25 **results in** failure to assemble the high-affinity IL-2 receptor (CD25 + CD122 + γc), abolishing high-affinity IL-2 binding ([PMID: 19348914](#)).

3. Loss of high-affinity IL-2 binding **leads to** deficient JAK1/JAK3 → STAT5 phosphorylation in T cells (*inferred from IL-2 receptor signaling biology*; reduced pSTAT5 is used functionally in diagnosis).
4. Deficient STAT5 signaling **results in** impaired FOXP3⁺ regulatory T-cell development, survival, stability, and suppressive function, and **impaired IL-10 production** by CD4 lymphocytes (PMID: 17196245, 19348914).
5. Treg failure **leads to** loss of peripheral immune tolerance, which **branches** into two co-existing outcomes:
6. **Branch A — Autoimmunity/lymphoproliferation:** uncontrolled activation and expansion of autoreactive effector T cells (notably CD8⁺STAT5⁺ cytotoxic T cells) **results in** multi-organ lymphocytic infiltration, autoimmune enteropathy, hepatitis, thyroiditis, cytopenias, and lymphadenopathy (PMID: 23416241, 9096364).
7. **Branch B — Immunodeficiency:** impaired antigen-specific T-cell responses **result in** susceptibility to recurrent viral, bacterial, and fungal infections (PMID: 23416241).
8. Combined tissue infiltration, autoimmune organ damage, malabsorptive enteropathy, and recurrent infection **lead to** failure to thrive and, untreated, early-childhood morbidity and mortality.



Mechanistic detail

- **Molecular pathways:** Interleukin-2 signaling via **JAK-STAT5** (Reactome R-HSA-451927 "Interleukin-2 signaling"; KEGG hsa04630 JAK-STAT; hsa04060 cytokine–cytokine receptor interaction). "Of crucial importance for the delivery of IL-2 signals to Treg cells is the expression of CD25, which, along with CD122 and gammac, confers high affinity binding to IL-2" ([PMID: 19348914](#)).
- **Cellular processes:** Failure of regulatory T-cell differentiation and suppression; unrestrained effector T-cell proliferation and inflammation.
- **Protein dysfunction:** Loss of function via truncation or conformational abrogation of surface expression ([PMID: 30742970](#)).
- **Immune involvement:** Simultaneous immunodeficiency and autoimmunity — a "Tregopathy." Caudy et al. showed a CD25-deficient patient had "defective IL-10 expression from CD4 lymphocytes, whereas a Foxp3-deficient patient expressed normal levels of

IL-10... although Foxp3 is not required for normal IL-10 expression by human CD4 lymphocytes, CD25 expression is important" (PMID: 17196245).

- **Tissue damage:** Direct cytotoxic and inflammatory injury — "Activated CD8(+)STAT5(+) T cells with lytic potential infiltrated the skin, even though FOXP3(+) Tregs were present" (PMID: 23416241).

Suggested GO / CL terms. Biological processes: interleukin-2-mediated signaling pathway (GO:0038110); regulatory T cell differentiation (GO:0045066); positive regulation of T cell proliferation (GO:0042102); JAK-STAT cascade (GO:0007259); tolerance induction (GO:0002507); negative regulation of immune response (GO:0050777). Cell types: regulatory T cell (CL:0000792); CD8-positive, alpha-beta T cell (CL:0000625); CD4-positive, alpha-beta T cell (CL:0000624); thymocyte (CL:0000893); B cell (CL:0000236).

7. Anatomical Structures Affected

Organ / body-system level. Immune/hematologic (primary), digestive, respiratory, integumentary, hepatobiliary, and endocrine systems. Sharfe et al. documented "extensive lymphocytic infiltration of tissues, including lung, liver, gut, and bone... accompanied by tissue atrophy and inflammation" (PMID: 9096364).

Structure	UBERON term
Intestine / gut	UBERON:0000160
Liver	UBERON:0002107
Lung	UBERON:0002048
Skin	UBERON:0002097
Bone / bone marrow	UBERON:0002371
Thymus	UBERON:0002370
Lymph node	UBERON:0000029
Spleen	UBERON:0002106
Thyroid gland	UBERON:0002046
Lacrimal/ocular surface & cornea	UBERON:0000964

Tissue / cell level. Primarily lymphoid tissue and infiltrated epithelial organs. Key cell types: FOXP3⁺ regulatory T cells (CL:0000792), CD8⁺ and CD4⁺ αβ T cells (CL:0000625, CL:0000624), thymocytes (CL:0000893), B cells (CL:0000236).

Subcellular level. Plasma membrane (GO:0005886) — site of the CD25 receptor; LOF variants prevent surface localization ([PMID: 30742970](#)). Also external side of plasma membrane (GO:0009897) and receptor complex (GO:0043235).

Localization / lateralization. Multi-organ and **bilateral/systemic** (e.g., bilateral ocular surface disease, generalized lymphoproliferation), not focal or unilateral.

8. Temporal Development

- **Onset:** Congenital/neonatal to early childhood, typically **insidious-to-subacute** with recurrent infections and chronic diarrhea in infancy.
- **Progression:** Chronic and **progressive** without treatment, with accumulating autoimmune organ damage and lymphoproliferation. Episodic autoimmune flares are superimposed.
- **Duration:** Chronic, lifelong; untreated cases carry high early-childhood mortality.

- **Remission:** No spontaneous remission. Durable remission/cure is **treatment-induced** by HSCT; sirolimus induces medical control.
 - **Critical period:** Early **molecular diagnosis and timely HSCT** represent the key window of therapeutic opportunity — "early clinical and molecular diagnosis... promptly led to HSCT, allowing complete resolution of the symptoms and definitive cure" ([PMID: 30742970](#)).
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9. Inheritance and Population

Epidemiology. Ultra-rare; ~13 molecularly confirmed cases reported worldwide as of 2026 ([PMID: 41659858](#)). **No population prevalence or incidence estimate is established** (data-level rarity).

Inheritance. **Autosomal recessive**, biallelic germline LOF variants — "a rare autosomal recessive inborn error of immunity" ([PMID: 41659858](#)). Both sexes affected (autosomal). Penetrance appears **complete** with biallelic LOF; expressivity is **variable** (spectrum from predominant enteropathy to lung, ocular, or hepatic involvement).

Consanguinity / founder effects. **Consanguinity is the principal population risk factor** — first-cousin unions are frequently reported ([PMID: 41694357](#)). No broad founder mutation established; variants are largely private.

Carrier frequency. Not established given rarity; heterozygous carriers are clinically unaffected.

Population demographics. Reported across multiple populations (Canada, Argentina, Italy, Morocco, others), with clustering in consanguineous families. No sex predilection. Age distribution is pediatric (presentation in infancy/early childhood). Data derive from **individual case reports**, not registries.

10. Diagnostics

Frontline immunophenotyping. Flow cytometry demonstrating absent/reduced surface CD25 on CD4⁺/activated T cells is the key rapid test: "Cytofluorimetric analysis revealed the total absence of CD25 cell surface expression and addressed IL2R α molecular investigation" (PMID: 30742970); "flow cytometry showed a complete absence of CD25 expression" (PMID: 41694357).

Molecular confirmation. Sequencing of *IL2RA* — single-gene testing, IEI/immune-dysregulation gene panels, or **whole-exome/whole-genome sequencing**. Genetic distinction from IPEX is essential: "Any patient with features of IPEX but with a normal Foxp3 gene should be screened for mutations in the IL-2 receptor subunit CD25" (PMID: 17196245).

Supporting laboratory tests. Lymphocyte subsets (variable; often normal-to-low T cells with expanded activated CD8⁺); immunoglobulins (often normal-to-high, hypergammaglobulinemia); autoantibodies (broad spectrum, including anti-mitochondrial antibodies; PMID: 20650610); impaired in-vitro T-cell proliferation to mitogens (PMID: 9096364, 23416241).

Functional test. Reduced IL-2-induced STAT5 phosphorylation in responder cells.

Biopsy/pathology. Affected-organ biopsy shows lymphocytic infiltration and tissue inflammation (e.g., follicular bronchiolitis with lymphocyte hyperplasia in lung).

Differential diagnosis. IPEX (FOXP3); CD122/IL2RB deficiency; STAT5b deficiency; LRBA deficiency; CTLA4 haploinsufficiency; other IPEX-like Tregopathies; and severe combined immunodeficiency (SCID). CD25 deficiency is distinguished from SCID by lymphoproliferation, autoimmunity, and preserved B-cell development (PMID: 24116927).

Screening. Not part of standard newborn screening. Cascade carrier and prenatal/preimplantation testing available once the familial variant is known.

11. Outcome / Prognosis

Untreated course. Severe and progressive, with high risk of death in early childhood from overwhelming infection and/or autoimmune organ damage — analogous to the lethal IL2/IL2R α -null mouse phenotype (death within ~5 weeks on the BALB/c background, PMID: 9065030) and severe multi-organ infiltration in humans (PMID: 9096364).

Treated course. HSCT is potentially curative — "complete resolution of the symptoms and definitive cure of the disease" after early transplantation ([PMID: 30742970](#)); long-term HSCT outcomes are reviewed in two additional novel cases ([PMID: 41659858](#)).

Morbidity / QoL. Substantial untreated morbidity (malnutrition, autoimmune organ damage, recurrent infection, dry-eye disease). Formal disability/QoL metrics are not available.

Prognostic factors. Timely molecular diagnosis and access to HSCT are the dominant prognostic determinants; extent of established autoimmune organ damage at transplant influences outcome. No validated prognostic biomarkers beyond the diagnostic markers.

12. Treatment

Definitive / curative therapy. Allogeneic hematopoietic stem cell transplantation (HSCT) replaces the defective hematopoietic compartment and restores functional Tregs — the only curative option (NCIT:C15431 Hematopoietic Stem Cell Transplantation) ([PMID: 30742970](#), [41659858](#)).

Bridging / medical immunomodulation. - **Rapamycin (sirolimus)** — mTOR inhibitor that spares/favors Tregs; reported as an "effective and safe treatment of a novel IL2RA deficiency" ([PMID: 31605764](#)) (NCIT:C1212 Sirolimus; CHEBI:9168). - **Corticosteroids and other immunosuppressants** for autoimmune flares. - **Immunoglobulin replacement and antimicrobial prophylaxis** for infection control. - **Nutritional support** for enteropathy and failure to thrive.

Pharmacogenomics. Not specifically defined for this disease.

Personalized / genotype-guided approach. Diagnosis-driven: confirmation of IL2RA LOF directs the patient toward definitive HSCT with sirolimus bridging, and enables family cascade testing.

Treatment strategy (algorithm). (1) Recognize IPEX-like phenotype → (2) flow cytometry for surface CD25 → (3) IL2RA sequencing → (4) initiate sirolimus/immunosuppression + supportive care to control autoimmunity and infection → (5) proceed to allogeneic HSCT as definitive cure.

13. Prevention

- **Primary prevention:** None possible for a Mendelian disorder (no modifiable exposure).
 - **Secondary/tertiary prevention:** Early molecular diagnosis, infection prophylaxis, immunosuppressive control of autoimmunity, and **timely HSCT** to prevent irreversible organ damage.
 - **Reproductive prevention / counseling:** Genetic counseling for autosomal recessive recurrence risk (**25% per pregnancy** for carrier couples); **cascade carrier testing** in consanguineous families; and **prenatal or preimplantation genetic testing** once the familial *IL2RA* variant is identified.
 - **Immunization / public-health interventions:** Not disease-specific, though routine infection prevention and prophylaxis are important for affected children.
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14. Other Species / Natural Disease

- **Taxonomy:** *Mus musculus* (NCBI Taxon 10090) is the principal model species. No described spontaneous natural companion-animal disease equivalent.
 - **Orthologous gene:** *Il2ra* (mouse ortholog of human *IL2RA*; MGI).
 - **Comparative biology:** Mouse *Il2ra*/*Il2* knockouts recapitulate lymphoproliferation, autoimmune hemolytic anemia, and inflammatory bowel disease due to defective Treg-mediated peripheral tolerance, confirming evolutionary conservation of the IL-2/CD25 tolerance mechanism ([PMID: 9065030](#)).
 - **Transmission / zoonotic potential:** None (genetic disorder).
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15. Model Organisms

Mammalian genetic models (mouse; NCBI Taxon 10090). - ***Il2*^{-/-} knockout mice:** On a mixed 129/Ola × C57BL/6 background, predominantly develop **ulcerative-colitis-like disease**; when backcrossed to **BALB/c** they develop **generalized autoimmune disease** — hemolytic anemia, follicular hyperplasia of lymphoid organs, and inflammation of pancreas, liver, heart, lungs, and thoracic vessels — dying within ~5 weeks, with uncontrolled polyclonal T- and B-cell activation and increased autoantibodies ([PMID: 9065030](#)). This demonstrates both phenotype

recapitulation and **genetic-background modifier effects**. - **Il2ra (CD25) knockout mice** (Willerford et al., *Immunity* 1995): massive lymphoproliferation, autoimmune hemolytic anemia, and inflammatory bowel disease from defective Treg-mediated tolerance — a direct genetic model of the human disorder. - **Foxp1/Foxp4 Treg-conditional knockouts**: combined deletion causes lymphoproliferation, inflammation, autoimmunity, and early lethality with reduced CD25 expression, mechanistically linking FOXP1/FOXP4 → Il2ra promoter → CD25 → Treg function ([PMID: 40794436](#)).

Model characteristics. Recapitulation is strong for **autoimmunity, lymphoproliferation, and colitis**; the immunodeficiency/infection-susceptibility axis is less emphasized in murine models. Genetic-background dependence is a key limitation for translating a single model to the full human phenotype spectrum.

Resources. MGI, IMPC/KOMP, IMSR (for Il2ra/Il2 alleles).

Evidence Base

PMID	Title (abbrev.)	Role / how it supports findings
9096364	Human immune disorder from IL-2Ra mutation (Sharfe 1997)	Original description; truncation of CD25, tissue lymphocytic infiltration (lung, liver, gut, bone)
17196245	CD25 deficiency causes IPEX-like syndrome, defective IL-10	Establishes IPEX-like designation, CD25-dependent IL-10; IL2RA screening if FOXP3 normal
23416241	Human IL2RA null mutation	CD8 ⁺ lymphoproliferation, impaired antigen responses, tissue infiltration despite FOXP3 ⁺ Tregs
19348914	IL-2/CD25 immunoregulatory mechanisms	CD25 + CD122 + γc confer high-affinity IL-2 binding to Tregs
30742970	New conformational mutation; HSCT cure	Conformational missense abrogates surface CD25; flow-cytometry diagnosis; HSCT cure
41694357	First Moroccan cases	AR inheritance, consanguinity, core phenotype, absent CD25 on flow
41659858	Two novel cases + review (long-term HSCT)	Confirms rarity (~13 cases), AR IEI definition, HSCT outcomes
24116927	Follicular bronchiolitis phenotype	Y41S missense; absent CD25/very low Tregs; lung disease; distinction from SCID
29252577	Severe dry eye in CD25 deficiency	Ocular surface disease; multi-tissue lymphocytic infiltration
31605764	Rapamycin treatment	Sirolimus as effective/safe bridging therapy
9065030	Il2 ^{-/-} BALB/c autoimmune mice	Mouse model; genetic-background modifier effect
40794436	FOXP1/FOXP4 in Tregs	FOXP1/4 bind Il2ra promoter, regulate CD25; modifier biology
20650610	Autoantibody spectrum in IPEX	Broad autoantibodies; pediatric PBC linked to CD25 deficiency
26106896	Fine-mapping IL2RA region	Common IL2RA variants as MS/T1D susceptibility (distinct from LOF disease)

PMID	Title (abbrev.)	Role / how it supports findings
21875375	10p15 IL2RA in T1D	rs41295061 association replicated in T1D
33193078	IL2RA in autoimmune thyroid disease	rs7093069 TT genotype increases AITD risk

Limitations and Knowledge Gaps

1. **Extreme rarity.** With only ~13–15 molecularly confirmed cases, all epidemiologic parameters (prevalence, incidence, carrier frequency, sex ratio, precise phenotype frequencies) are unquantified; phenotype frequencies remain qualitative.
2. **Case-report evidence base.** Nearly all human data come from single cases/small series with inherent selection and publication bias. No natural-history cohort or registry exists.
3. **Genotype–phenotype correlation** is undefined given so few variants.
4. **Mechanistic steps partly inferred.** The STAT5-phosphorylation step and gut-microbial drive of enteropathy are inferred from IL-2 biology and mouse data rather than directly demonstrated in every patient.
5. **Modifier genes in humans** are unknown; mouse background effects suggest they exist.
6. **No formal QoL/disability metrics** or standardized outcome measures are available.
7. **Long-term HSCT outcomes** are based on very few transplanted patients.

Proposed Follow-up Experiments / Actions

1. **Establish an international IL2RA-deficiency patient registry** (via IEI consortia such as ESID/USIDNET) to aggregate genotype, phenotype, treatment, and outcome data.
2. **Systematic functional characterization** of each reported IL2RA variant (surface expression, IL-2 binding, pSTAT5 response) to build a variant-effect map.
3. **Single-cell / immune profiling** of patient PBMCs and affected tissues to resolve Treg vs. effector-cell dynamics and IL-10 deficits at cellular resolution.
4. **Standardized HSCT outcome tracking** (conditioning regimen, chimerism, autoimmunity resolution, long-term immune reconstitution).

5. **Evaluate low-dose IL-2 and sirolimus** as bridging therapies in a prospective, protocolized fashion (noting that high-affinity receptor loss may blunt low-dose IL-2 efficacy — a hypothesis worth testing).
 6. **Explore gene-correction / gene-editing** (e.g., autologous HSC IL2RA correction) as a future curative alternative to allogeneic HSCT.
 7. **Carrier screening in consanguineous populations** where cases have clustered, to enable earlier diagnosis and reproductive counseling.
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Evidence source types are indicated throughout: human clinical (case reports/series), model organism (mouse knockouts), and in vitro/functional (flow cytometry, STAT5 assays). Ontology suggestions (HPO, GO, CL, UBERON, CHEBI, NCIT, MONDO) are provided inline for knowledge-base ingestion.
